Jonas Ngnawé

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Google Scholar • Stanford Profile

Education

Université Laval (IID), Mila - Quebec AI Institute, Stanford (VSR, 2025-2026) **Quebec, Canada**
PhD in Computer Science: Adversarial Machine Learning *2021–2026*

AIMS African Master's of Machine Intelligence **Kigali, Rwanda**
MSc in Machine Learning *2018–2019*
Funded by Meta and Google; taught by world top researchers.

African Institute for Mathematical Sciences (AIMS) **Limbe, Cameroon**
MSc in Mathematical Sciences *2016–2017*

École Polytechnique **Yaoundé, Cameroon**
MSc in Computer Engineering *2011–2016*

Summer Schools: Machine Learning Summer School (MLSS), London 2019; Deep Learning & Reinforcement Learning Summer School (DLRL), Edmonton 2025.

Publications

Robust Fine-Tuning from Non-Robust Pretrained Models: Mitigating Suboptimal Transfer with Epsilon-Scheduling, **Ngnawé, J.**, Heuillet, M., Sahoo, S., Pequignot, Y., Ahmad, O., Durand, A., Précioso, F., & Gagné, C. (NeurIPS 2025 Workshop on Reliable ML, ICLR 2026). [arXiv]

Detecting Brittle Decisions for Free: Leveraging Margin Consistency in Deep Robust Classifiers, **Ngnawé, J.**, Sahoo, S., Pequignot, Y., Précioso, F., & Gagné, C. (NeurIPS 2024). [NeurIPS Proceedings]

A Layer Selection Approach to Test Time Adaptation, Sahoo, S., ElAraby, M., **Ngnawé, J.**, Pequignot, Y., Précioso, F., & Gagné, C. (AAAI 2025). [AAAI Proceedings]

A Guide to Robust Generalization: The Impact of Architecture, Pre-training, and Optimization Strategy, Heuillet, M., Bhagwatkar, R., **Ngnawé, J.**, Pequignot, Y., Larouche, A., Gagné, C., Rish, I., Ahmad, O., & Durand, A. (NeurIPS 2025 Workshop on Reliable ML). [arXiv]

Robustmix: Improving Robustness by Regularizing the Frequency Bias of Deep Networks, **Ngnawé, J.**, Njifon, M. A., Heek, J., & Dauphin, Y. (NeurIPS 2022 Workshop on Distribution Shifts). [OpenReview]

Scalable Inference with Distributed Gaussian Processes using the Wasserstein Barycenter, **Ngnawé, J.**, Deisenroth, M. (Master's project). [GitHub]

k-Clustering Problem: Application to a Field Technician Allocation Optimization Problem, **Ngnawé, J.**, Zacko, E., Wandji, N., & Ndjinga, M. (Master's project). [ResearchGate]

AI/ML Work Experience

Stanford Artificial Intelligence Lab (SAIL) **Palo Alto, USA**
Visiting Student Researcher *Oct 2021 – to Date*

- Adversarial robustness of Vision-Language Models.
- AI Measurement Science applied to LLMs Evaluation

Google Brain **Accra, Ghana**
Google AI Resident *Oct 2019 – Sep 2021*

- Development of robust methods for deep neural networks.
- Compression of deep neural networks by quantization and pruning.

Other Work Experience

GroupOne Holding Company **Limbe, Cameroon**
Computer Engineer *Jan – Aug 2018*

- Website development; fuel consumption prediction projects.
- Studies for the creation of a startup incubator (business plan).
- Industrial maintenance (refinery) of the Honeywell Experion System; IT support.

Tools: HTML5/CSS3/JavaScript/jQuery/jQueryUI/Ajax/PHP, MySQL, PMI Methodology, Business Model Canvas.

Cameroon
Freelance Software Developer *2017–2018*

- Digitization of administrative procedures for various ministries in Cameroon.
- Development and maintenance of WordPress plugins (WooCommerce).

Tools: DevOps, Electronic Document Management (EDM)/Electronic Archiving (Maarch RM), PHP, WooCommerce, MySQL Workbench, Microsoft Access, Git/GitLab/GitHub, Java, NetBeans, SCRUM.

Bolloré Africa Logistics **Douala, Cameroon**
Software Developer Intern *Feb – Aug 2016*

- Automation and analysis of the review of access rights to the Information System.
- Detection of abnormal access rights.

Tools: Talend Open Studio, QlikView, C#, ASP.NET, Microsoft SQL Server, Visual Studio, UML/Merise, BPMN 2.0, PMI Methodology.

Talks

Host at Kmer AI Talks.

2024: Panelist, "AI Regulation & Fairness in the Generative AI Era," Black in AI Workshop @ NeurIPS.

2024–2025: "Detecting Brittle Decisions for Free" — Kmer AI (Jun 2024); Galsen AI (Jan 2025).

2022: Keynote, "Robustness in AI," Deep Learning IndabaX-Cameroon (Dec 27, 2022).

Reviews

NeurIPS, ICLR, AAAI; IEEE Transactions on Image Processing.

Algorithmic Fairness workshops @ NeurIPS 2021, 2022, 2023, 2024.

Montreal AI Symposium (2022, 2024); Black in AI Workshop (2022–2024).

Deep Learning Indaba, Pan-African Machine Learning Conference (2023).

Leadership & Event Organization

Organizing committee: Interpolate Workshop @ NeurIPS 2022 (First Workshop on Interpolation Regularizers and Beyond).

Co-organizer: Machine Learning Projects Competition (MLPC) 2020–2022, Polytechnique Yaoundé.

Co-organizer: Two-week introductory course in Machine Learning (Sep 2019), Polytechnique Yaoundé.

Co-organizer: First Deep Learning IndabaX Rwanda (Apr 2019), Kigali, Rwanda.

Co-Founder of Kmer-AI, an association for promoting and popularizing AI and ML in Cameroon (since 2020).

AI/ML Skills

ML Frameworks: PyTorch, TensorFlow, Scikit-Learn

MLOps Tools: Git/GitHub, Weights & Biases, Docker, Ray, Slurm, Google Cloud Platform

Other: Distributed Training, VLMs/LLMs, Adversarial Evaluation